AWAS: Smart Water Management and Billing System

System Evaluation Questionnaire based on ISO/IEC 25010 · Barangay Adlay

Dear Respondent,

We are evaluating AWAS, a system for meter reading, water billing, online payment (GCash), membership applications and notifications. Please rate each statement based on your experience using the system. Your answers are confidential and will be used for academic purposes only.

Instructions

1. Fill in your profile in Part I.
2. In Part II, rate each statement from 1 to 5 by putting a check (✓) in the column that matches your answer.
3. Answer every statement. Only IT experts or developers answer Section G (Maintainability); other respondents may skip it.
4. Write any comments or suggestions in Part III.

Part I. Respondent Profile

Name (optional)	
Purok	
Age	
Sex	☐ Male ☐ Female
Role	☐ Administrator / Staff ☐ Meter Reader ☐ Consumer ☐ IT Expert / Developer
Computer / smartphone use	☐ Daily ☐ A few times a week ☐ Rarely
Device used for AWAS	☐ Desktop / Laptop ☐ Smartphone ☐ Tablet

Rating Scale

Rating	Description	Meaning
5	Strongly Agree	The system fully meets the statement
4	Agree	The system meets the statement with minor issues
3	Neutral	The system partly meets the statement
2	Disagree	The system rarely meets the statement
1	Strongly Disagree	The system does not meet the statement

Part II. System Evaluation (ISO/IEC 25010)

#	Statement	5	4	3	2	1
A. Functional Suitability — *The degree to which AWAS provides functions that meet stated and implied needs.*						
1	**Completeness:** The system provides all the functions needed for water billing: consumer records, meter reading, bill generation, payment recording and reports.					
2	**Completeness:** The system supports online membership application and approval.					
3	**Correctness:** The system computes water bills accurately based on meter readings and billing rates.					
4	**Correctness:** The system shows the correct balance, payment history and unpaid bills of each consumer.					
5	**Appropriateness:** The functions help users finish tasks such as recording payments and viewing bills easily.					
B. Performance Efficiency — *The performance of AWAS relative to the resources it uses.*						
1	**Time behavior:** Pages load within a few seconds.					
2	**Time behavior:** Searching consumers, generating bills and producing reports are fast.					
3	**Time behavior:** New meter readings and notifications appear without long delays.					
4	**Resource utilization:** The system runs smoothly without slowing down the device.					
5	**Capacity:** The system handles many consumers, bills and simultaneous users without problems.					
C. Compatibility — *The degree to which AWAS works with other systems and environments.*						
1	**Co-existence:** The system runs alongside other applications on the same device without conflict.					
2	**Interoperability:** The system works properly with online payment (GCash) and email notifications.					
3	**Interoperability:** Bills and receipts can be printed or downloaded correctly.					
4	**Interoperability:** Reports exported from the system can be opened in other programs.					
D. Usability — *The degree to which AWAS can be used effectively, efficiently and with satisfaction.*						
1	**Appropriateness recognizability:** I can easily tell whether the system fits my needs.					
2	**Learnability:** The system is easy to learn even without training.					
3	**Operability:** Menus, buttons and forms are easy to use and control.					
4	**User error protection:** The system prevents mistakes and shows clear error messages.					
5	**User interface aesthetics:** The layout, colors and fonts are pleasing and consistent.					
6	**Accessibility:** The system is easy to use on both computers and smartphones.					

#	Statement	5	4	3	2	1
E. Reliability — *The degree to which AWAS performs its functions without failure over time.*						
1	**Maturity:** The system works without errors or crashes during normal use.					
2	**Availability:** The system is accessible whenever I need it.					
3	**Fault tolerance:** The system keeps working even when wrong inputs or connection problems occur.					
4	**Recoverability:** Data such as readings, bills and payments are not lost after an error or interruption.					
F. Security — *The degree to which AWAS protects information and data.*						
1	**Confidentiality:** Only authorized users can view consumer records, bills and payments.					
2	**Integrity:** Bills, readings and payment records cannot be changed without authorization.					
3	**Non-repudiation:** Each payment and transaction has a receipt or record as proof.					
4	**Accountability:** Actions are linked to the user account that performed them.					
5	**Authenticity:** The system verifies users through login, email verification and role-based access (Admin, Meter Reader, Consumer).					
G. Maintainability — *The degree to which AWAS can be modified and improved efficiently. (For IT experts / developers only.)*						
1	**Modularity:** The system is divided into modules, so changing one does not affect the others.					
2	**Reusability:** Components of the system can be reused in other features or systems.					
3	**Analysability:** Errors and their causes are easy to find.					
4	**Modifiability:** Billing rates, puroks, settings and FAQs can be updated without changing the code.					
5	**Testability:** The functions of the system can be easily tested.					
H. Portability — *The degree to which AWAS can be transferred to other devices or environments.*						
1	**Adaptability:** The system works on different browsers (Chrome, Edge, Firefox) and screen sizes.					
2	**Installability:** The system can be installed or deployed easily on a new server or computer.					
3	**Replaceability:** The system can replace the current manual or old billing process.					

Part III. Comments and Suggestions

1. What features of AWAS did you find most useful?

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_____

2. What problems did you experience while using the system?

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_____

3. What improvements or additional features would you suggest?
_____
_____

Respondent's signature:	**Date:**
_____	_____

Thank you for your time and cooperation.